Scale Factor: Perimeter and Area

Teacher's Edition

The Period

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

Must Find the perimeter and area of a scale copy from the original and k (perimeter × k, area × k^2), and find k from two areas when the ratio is a perfect square. (MA.7.GR.1.5, clarification 2.)

Must not Area ratios that are not perfect squares (an approximate root); the area of a figure the class has no formula for; area and perimeter in different units (5.04 and 5.08–09).

Target I can find the perimeter and the area of a scale copy from the original and k, and find k from two areas.

MTR **MTR.2.1** — Notes II: the same copy as a grid picture, as 15 → 60, and as $k^2 = 4$ · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote on k against k^2 · **MTR.5.1** — boards 1–2: the same k on the perimeter, then the area, and the room is asked why the second answer is bigger

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	10 min
Example 1	8–9	4 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. The perimeter is lengths, so it is multiplied by k.
2. Area is two lengths multiplied, so it is multiplied by k twice: k squared.
3. From two areas, k is the square root of the ratio.

Slide by Slide

1. **Scale Factor: Perimeter and Area.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the figure Notes I and II scale: 3 by 5, P = 16, A = 15. *1 retrieves 5.01 — copy ÷ original; k under 1 (yesterday's closure promised it) (yesterday). 2 retrieves 4.05 — a perfect square's root; Notes III finds k this way (last week). 3 retrieves 3.01 — powers; k^2 is a power (last unit). 4 retrieves MA.6.GR.2.1 — perimeter is the sum of the sides, area is length × width; today both are scaled (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. k = 0.4 2. 12 3. 9 and 27 4. P = 16 cm, A = 15 cm^2
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the figure and the two lines.** The 3 by 5 from the warm-up at k = 2: 6 by 10.
6. **Notes.** Area is length × width, and both were multiplied by k, so the area is multiplied by k twice: 15 → 60, four times as much, not twice.
7. **Notes.** Run it backwards.
8. **Example 1 — Perimeter and area from k.** Ask which rule each part uses before anyone multiplies: perimeter × k, area × k^2.
9. **Example 1 — Worked.** P = 12 → 36 (× 3); A = 8 → 72 (× 9).
10. **Your Turn — Same steps, your numbers. Boards up when done.** P 18 → 72, A 18 → 288.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — k from two areas, then a side.** Ask what 180 ÷ 20 is, and whether that number is k.
13. **Example 2 — Worked.** 240 ÷ 15 = 16 is k^2, so k = 4, and the side is 12.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 300 ÷ 12 = 25, k = 5, the side is 10.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. A figure with perimeter 14 cm is scaled by k = 2. What is the copy's perimeter? → ***28 cm*** *56 — perimeter × k^2 [Notes I: a perimeter is lengths, × k]; 16 — added k [7.GR.1.5: the additive error — Math Nation's named misconception]*

18–19. Board 2. A figure with area 12 cm^2 is scaled by k = 2. What is the copy's area? → ***48 cm^2*** *24 — area × k [B1G-M 7.GR.1.5 named misconception: scaling area like length]*

20–21. Board 3. A figure with area 48 cm^2 is scaled by k = ½. What is the copy's area? → ***12 cm^2*** *24 — area × k [7.GR.1.5: area × k^2]; 96 — divided by ½ as if k were 2 [5.01: k under 1 shrinks]*

22–23. Board 4. A figure with area 10 cm^2 is scaled by k = 3. What is the copy's area? → ***A — 90 cm^2*** *(B) area × 2k: 'two lengths, so twice k' [Notes II: k × k, not k + k]; (C) area × k: the guide's named misconception [B1G-M 7.GR.1.5: scaling area like length]; (D) 10 + 3: the additive error [7.GR.1.5: Math Nation's named misconception] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

24–25. Board 5. A figure with area 8 cm² is scaled by k = 1.5. What is the copy's area? → ***18 cm²*** *12 — area × k [7.GR.1.5]; 24 — area × 2k [Notes II: k × k]*

26–27. Board 6. A figure's perimeter is 20 cm; its copy's perimeter is 60 cm. What is the area ratio, copy to original? → ***D — 9*** *(A) 2k: 'area has two dimensions, so double it' [Notes II]; (B) the perimeter ratio handed in as the area ratio [Notes I and II: perimeter gives k, area is k²]; (C) 60 − 20: the additive error [7.GR.1.5: Math Nation's named misconception] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. A triangle with sides 6, 8 and 10 cm is scaled by k = ½. Its area is 24 cm². Find the copy's perimeter and area. → ***P = 12 cm, A = 6 cm²*** *P 12, A 12 — the area halved like a length [7.GR.1.5]; P 6, A 6 — both quartered*

30–31. Board 8. A 4 in by 6 in photo, taken 3 years ago, is enlarged at k = 2.5. What is the enlargement's area? → ***150 in²*** *The 3 years is not needed. 60 — area × k [7.GR.1.5]; 120 — 24 × 5, k doubled [Notes II]*

32–33. Board 9. Jo says that doubling every side of a rectangle doubles its area. Explain why not, with a counterexample. → ***No: 2 by 7 doubled is 4 by 14, area 14 → 56 — × 4, as both lengths doubled.*** *'Yes, because every side doubles' — scaling area like length [B1G-M 7.GR.1.5]; 'No, it is k²' with no numbers — the rule without the reason*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–2 keep k = 2 and move from perimeter to area; 3 takes k under 1; 4 asks the area as a vote; 5 makes k a decimal. The pattern breaks at 6, where k comes from perimeters and the area ratio is asked. 7 is a triangle, 8 the extra number, 9 the reason in writing. The independent set walks the same path: perimeter × k, area × k², k under 1, both on one rectangle, k from areas, the area ratio from perimeters.

Misconceptions to Watch

Board 1. 56 — perimeter × k² [Notes I: a perimeter is lengths, × k]

Board 2. 24 — area × k [B1G-M 7.GR.1.5 named misconception: scaling area like length]

Board 3. 24 — area × k [7.GR.1.5: area × k²]

Board 4. area × 2k: 'two lengths, so twice k' [Notes II: k × k, not k + k]

Board 5. 12 — area × k [7.GR.1.5]

Board 6. 2k: 'area has two dimensions, so double it' [Notes II]

Board 7. P 12, A 12 — the area halved like a length [7.GR.1.5]

Board 8. 60 — area × k [7.GR.1.5]

Board 9. 'Yes, because every side doubles' — scaling area like length [B1G-M 7.GR.1.5]

Materials Whiteboards and markers. A sheet of grid paper under the document camera for Notes II's four-copies picture, drawn live if the slide's grid is not enough. Calculators are allowed on everything and printed on nothing.